Khai-Nguyen Nguyen

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Education

College of William and Mary — Virginia, USA
Ph.D. Student in Computer Science — 2023 - present
- **GPA:** 3.96/4.00
- **Advisor:** Dr. Antonio Mastropaolo

Bucknell University — Pennsylvania, USA
B.Sc. in Computer Science and Engineering, minor in Statistics — 2019 - 2023
- **GPA:** 3.94/4.00 *(Summa Cum Laude)*

Research Interests

Natural Language Processing, Trustworthy and Explainable AI, Multimodal AI, AI for Healthcare

Selected Publications

♣: *equal contribution*

1. **Vision Language Models are Biased** [pdf]
   An Vo♣, **Khai-Nguyen Nguyen♣**, Mohammad Reza Taesiri, Vy Tuong Dang, Anh Totti Nguyen, Daeyoung Kim
   ICLR 2026
2. **Sentiment Reasoning for Healthcare** [pdf]
   Khai-Nguyen Nguyen♣, Khai Le-Duc♣, Bach Phan Tat, Duy Le, Truong-Son Hy
   ACL 2025, Industry Track (Oral)
3. **Medical Spoken Named Entity Recognition** [pdf]
   Khai Le-Duc, David Thulke, Hung-Phong Tran, Long Vo-Dang, **Khai-Nguyen Nguyen**, Truong-Son Hy, Ralf Schluter
   NAACL 2025, Industry Track
4. **Real-time Speech Summarization for Medical Conversations** [pdf]
   Khai Le-Duc♣, **Khai-Nguyen Nguyen♣**, Long Vo-Dang, Truong-Son Hy
   Interspeech 2024 (Oral)
5. **Getting away with network pruning: From sparsity to geometry and linear regions** [pdf]
   Jeffrey Cai♣, **Khai-Nguyen Nguyen♣**, Nishant Shrestha, Aidan Good, Ruisen Tu, Xin Yu, Shandian Zhe, Thiago Serra
   CPAIOR 2023
   Sparsity in Neural Networks Workshop @ICLR 2023

Preprints

♣: *equal contribution*

1. **Toward Explaining Large Language Models in Software Engineering Tasks** [pdf]
   Antonio Vitale♣, **Khai-Nguyen Nguyen♣**, Denys Poshyvanyk, Rocco Oliveto, Simone Scalabrino, Antonio Mastropaolo
   Submitted to TOSEM 2025
2. **S-Chain: Structured Visual Chain-of-Thought for Medicine** [pdf]
   Khai Le-Duc, Phuong T.H. Trinh, Duy Minh Ho Nguyen, Tien-Phat Nguyen, Nghiem Tuong Diep, An Ngo, Tung Vu, Trinh Vuong, Anh-Tien Nguyen, Nguyen Dinh Mau, Van Trung Hoang, **Khai-Nguyen Nguyen**, Hy Nguyen, Chris Ngo, Anji Liu, Nhat Ho, Anne-Christin Hauschild, Khanh Xuan Nguyen, Thanh Nguyen-Tang, Pengtao Xie, Daniel Sonntag, James Zou, Mathias Niepert, Anh Totti Nguyen
   arXiv 2025

Research Experiences

AURA Lab, College of William and Mary, Virginia, USA

Research Assistant. Advisor: Prof. Antonio Mastropaolo 2024.10 - present

Paper: *Toward Explaining Large Language Models in Software Engineering Tasks*

- Implemented a model-agnostic SHAP-based framework to quantify the effect of each input feature of software engineering prompts on the model output
- Proposed and conducted quantitative experiments to understand the attribution capabilities of our framework compared to random and LLM baselines

Paper: *Resource-Efficient & Effective Code Summarization*

- Built a PEFT training pipeline for code language models (e.g., `DeepSeek-Coder`, `CodeLlama`) to quantify the impact of QLoRA on code summarization

Nguyen Lab, Auburn University, Alabama, USA

Remote Collaborator. Advisor: Prof. Anh Totti Nguyen 2025.01 - present

Paper: *Vision Language Models are Biased*

- Conducting knowledge editing experiments on VLMs to mitigate knowledge bias using the `EasyEdit` package
- Conducting ablation and activation patching experiments on open-source VLMs using the `nnsight` package to locate the source of bias in the model
- Built an AI-assisted image generation pipeline for realistic-looking counterfactual images (e.g., animals, logos and flags) using `Gemini-2.0-Flash`
- Conducted ablation experiments (e.g., linear probing, attention visualization, image background removal with `LangSAM`) to locate contributing factors to knowledge bias

FPT Software AI Center, Vietnam

Remote Research Intern. Supervisor: Prof. Truong-Son Hy 2024.01 - 2024.09

Paper: *Sentiment Reasoning for Healthcare*

- Proposed the *sentiment reasoning* task and implemented a chain-of-thought finetuning pipeline for LLMs in medical sentiment analysis
- Co-led the construction of the `Sentiment Reasoning` dataset consisting of 30,000 samples across 5 different languages

Paper: *Real-time Speech Summarization for Medical Conversations*

- Co-led the construction of the `VietMed-Sum` dataset for medical speech summarization consisting of 24,000 samples and built its training pipeline
- Proposed and conducted an ablation study on the effect of mixing human-curated and synthetic data for medical summarization in budget-constrained scenarios

Analytics Lab, Bucknell University, Pennsylvania, USA

Undergraduate Research Assistant. Advisor: Prof. Thiago Serra 2022.05 - 2023.01

Paper: *Getting away with network pruning: From sparsity to geometry and linear regions*

- Co-developed a theorem establishing an upper bound on the expressiveness, quantified by linear regions, of ReLU-based deep neural networks.
- Implemented `GurobiPy`-based functions that identify stably active and inactive neurons, and count the linear regions in deep neural networks

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Industry Experiences

Machine Learning Intern, *CodaMetrix* | Boston, USA 2025.05 - 2025.08

- Developed a Gemini-based judge to evaluate and refine medical entity annotations, improving performance by 2–4% in F1 over human baseline
- Designed and deployed entity extraction LLMs with DSPy prompt optimization on Databricks, achieving an overall F1 of 70% on `Llama-3.1-70B`

Machine Learning Intern, *CodaMetrix* | Boston, USA 2024.05 - 2024.08

- Trained a BERT-based LLM for ICD-10 multilabel classification, outperforming the deployed system by 4% in accuracy
- Accelerate training by 4x on Databricks using distributed and parallel training

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Presentations

Sentiment Reasoning for Healthcare
Oral presentation at ACL 2025 Industry Track

Medical Spoken Named Entity Recognition
Poster presentation at NAACL 2025 Industry Track

Real-time Speech Summarization for Medical Conversations
Oral presentation at Interspeech 2024 & Poster presentation at MASC-SLL 2024

Awards and Honors

- **Recipient**, Computer Science Fellowship, College of William and Mary — 2023
  Awarded $500 by the CSCI admissions committee
- **Recipient**, Bucknell University Grant — 2019 - 2023
  Awarded approximately $60,000 annually
- **Dean's List**, Bucknell University — 2019 - 2023
  Achieved a GPA of 3.6 or higher every semester
- **Recipient**, Program for Undergraduate Research Grant, Bucknell University — 2021
  Awarded $4,000 for summer research
- **Honorable Mention**, Mathematical Contest in Modeling — 2020
  Ranked top 25% among 13753 participating teams

Teaching Experiences

William and Mary: CSCI 140: Intro to Data Science
CSCI 304: Computer Organization
CSCI 435/535: Software Engineering

Bucknell University: CSCI 204: Data Structures and Algorithms
MATH 201, 202: Calculus I, II

Mentorship

- **An Ngo**: Bucknell University — 2025.01 - 2025.09
- **Linh Nguyen**: Bucknell University → SDE@Amazon — 2024.09 - 2025.01
- **Duy Le**: Bucknell University — 2024.05 - 2024.08
- **Hung Ngo**: Bucknell University → R&D Engineer@Medivis — 2023.05 - 2023.09

Services

Bucknell's Machine Learning Association | Co-founder and Secretary — 2022 - 2023
- Connected machine learning enthusiasts at Bucknell by inviting guest speakers from both industry and academia and organizing a machine learning-themed hackathon

Vietnamese Student Association at Bucknell | Vice President of Finance — 2021 – 2022
- Organized traditional Vietnamese events such as Mid-Autumn Festival and Lunar New Year Festival, managed the club budget for said events

Selected Skills

Programming Languages: Python, Java, C/C++, R, SQL
Libraries and Frameworks: PyTorch, TensorFlow, Pandas, HuggingFace, vLLM, nnsight, TransformerLens, SentenceTransformers, DSPy, Matplotlib, Seaborn, GurobiPy